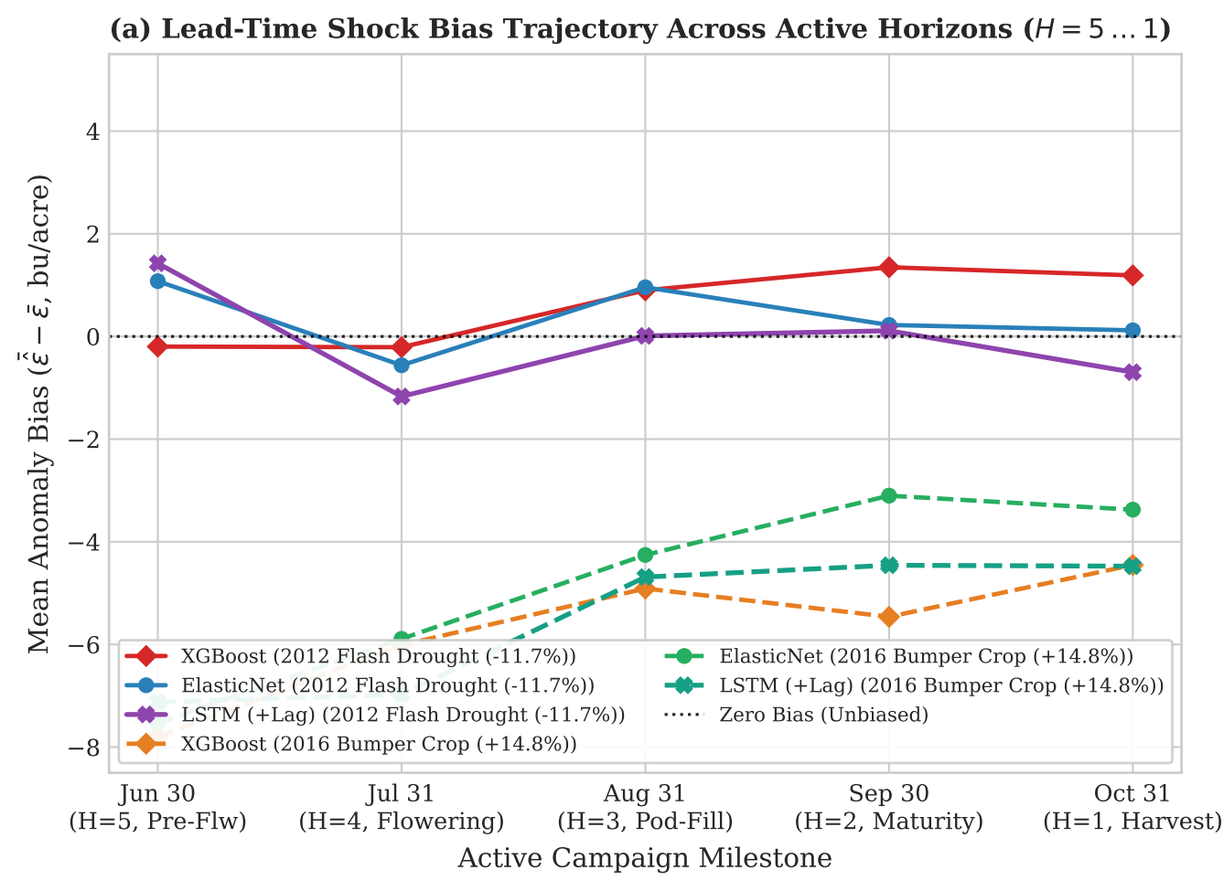
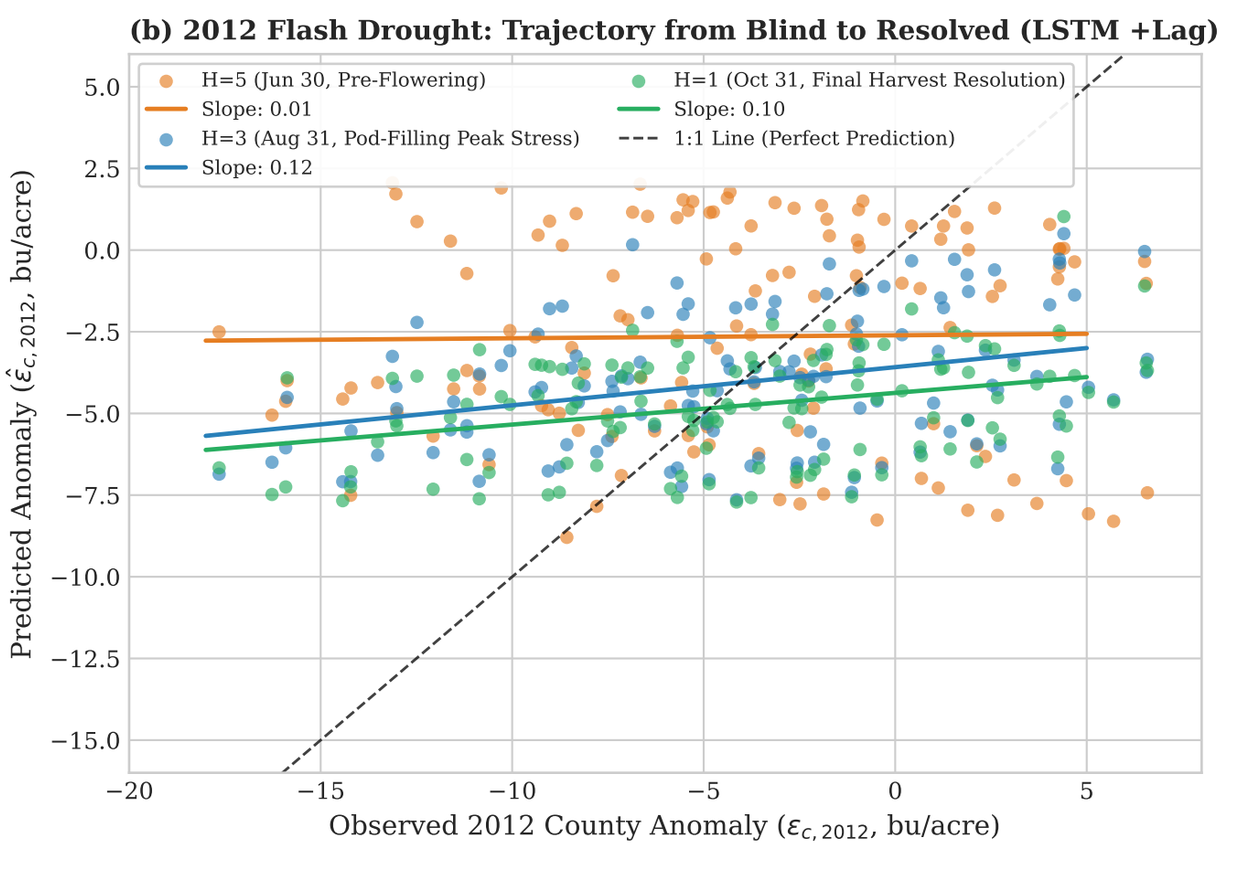

(a) Lead-Time Shock Bias Trajectory Across Active Horizons ($H = 5 \dots 1$)
Mean Anomaly Bias ($\bar{\hat{\varepsilon}} - \bar{\varepsilon}$, bu/acre)
4
2
0
−2
−4
−6
−8
XGBoost (2012 Flash Drought (-11.7%))
ElasticNet (2012 Flash Drought (-11.7%))
LSTM (+Lag) (2012 Flash Drought (-11.7%))
XGBoost (2016 Bumper Crop (+14.8%))
ElasticNet (2016 Bumper Crop (+14.8%))
LSTM (+Lag) (2016 Bumper Crop (+14.8%))
Zero Bias (Unbiased)
Jun 30
(H=5, Pre-Flw)
Jul 31
(H=4, Flowering)
Aug 31
(H=3, Pod-Fill)
Sep 30
(H=2, Maturity)
Oct 31
(H=1, Harvest)
Active Campaign Milestone


(b) 2012 Flash Drought: Trajectory from Blind to Resolved (LSTM +Lag)
H=5 (Jun 30, Pre-Flowering)
Slope: 0.01
H=3 (Aug 31, Pod-Filling Peak Stress)
Slope: 0.12
H=1 (Oct 31, Final Harvest Resolution)
Slope: 0.10
1:1 Line (Perfect Prediction)
Predicted Anomaly ($\hat{\varepsilon}_{c,2012}$, bu/acre)
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−20
−15
−10
−5
0
5
Observed 2012 County Anomaly ($\varepsilon_{c,2012}$, bu/acre)